

Why Do Some Regions Remain Poor?

Geography vs. Historical Infrastructure:
A Global Analysis with India as a Case Study

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Tools Used: Python | SQL | Power BI | World Bank WDI

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Abstract

This study explores one of the most important questions in development economics: why do some regions of the world remain poor while others become wealthy? Two main explanations have been debated by economists for decades: geography and infrastructure. Geography refers to natural features such as whether a country is landlocked, how far it is from the coast, and how difficult its terrain is. Infrastructure refers to things that humans build, such as railways, roads, and access to electricity.

To answer this question, I analyzed data from over 180 countries using the World Bank's World Development Indicators (WDI) dataset for 2000–2022. I then conducted a deeper case study of India using district-level census data and historical railway network maps to understand how British colonial infrastructure decisions from the 1800s might still be affecting regional development today.

The main findings of this study are as follows: First, access to electricity is almost perfectly correlated with GDP per capita; every country with 100% electricity access has a GDP above USD 50,000 per person. Second, the rail network size is strongly associated with economic development. Third, India's GDP grew by 177% between 2000 and 2022, but significant inequality remains between well-and poorly connected districts. These findings suggest that infrastructure, not just geography, plays a critical role in explaining why some regions remain poor.

1. Introduction

Why some countries are much richer than others? Why does a person born in Norway earn 300 times more in their lifetime than a person born in Burundi? Is it because of their geographical location on the map? Or is it because of what their country built over time — railways, roads, and power lines?

This is not only an interesting question. It is one of the most studied and debated topics in the field of economics. This question is important because if geography is the main cause of poverty, there is very little we can do to fix it. However, if infrastructure is the main cause, building more roads, railways, and power grids could be the solution.

I chose this topic because I wanted to go beyond merely analyzing numbers and attempt to answer a meaningful real-world question using data. As a first-year data analytics student, this project provided me with the opportunity to work with real datasets from the World Bank and the Indian Census, clean and process them using Python, store them in a SQL database, and visualize them using Power BI.

This study is divided into two parts. The first part looks at the global picture — across 180+ countries, does geography or infrastructure better explain differences in GDP per capita and poverty rates? The second part zooms into India — within India's 676 districts, does the presence of railway lines (many of which were built during British colonial rule) explain why some districts are more developed than others?

Research Questions
1. Does geography or infrastructure better predict GDP per capita across countries globally?
2. Does access to electricity correlate with lower poverty rates?
3. Do districts with more railway access show higher levels of development in India?
4. How has India's GDP per capita changed over time compared with its neighbors?

2. Literature Review

Several economists have studied this question. Their work helped me design my analysis.

2.1 The Geography Argument

Jeffrey Sachs, a famous development economist at Columbia University, argued that geography plays a huge role in development. Countries near the equator tend to have more tropical diseases, such as malaria, which reduces worker productivity. Landlocked countries cannot access sea trade routes, which makes them poorer on average. Sachs found that being landlocked reduces a country's GDP by approximately 30%.

Similarly, Gallup, Sachs, and Mellinger (1999) showed that countries with access to coastlines and navigable rivers have historically been much richer because they could trade goods more easily. In this view, geography is destiny.

2.2 The Infrastructure Argument

However, other economists disagree with this view. Daron Acemoglu and James Robinson, in their famous book 'Why Nations Fail' (2012), argue that institutions and infrastructure matter far more than geography. Countries that invested in railways, education, and legal systems became rich, regardless of their location.

The most important study for my India case study is Dave Donaldson's research paper, 'Railroads of the Raj' (2018), which won the John Bates Clark Medal. Donaldson studied British colonial India and found that districts connected to the railway network saw significant increases in agricultural income and trade. The effects of railway access persisted for decades after the country's independence.

2.3 My Contribution

While these researchers used very complex statistical methods, my goal in this study is much simpler. I want to use the data skills I have learned — Python, SQL, and Power BI — to explore these same questions in a visual and accessible way. I do not claim to have definitively solved this debate. Instead, I want to show what the patterns in the data look like and let those patterns guide our discussion.

3. Data and Methodology

3.1 Datasets Used

Dataset	Source	What it contains	Used for
World Bank WDI	World Bank Open Data	GDP per capita, poverty rate, electricity access, rail lines for 180+ countries 2000–2022	Global analysis
India Census 2011	Census of India / Kaggle	District-level housing, electricity, water access across 640 districts	India deep dive
India Railway Network	OpenStreetMap via HDX	Location of all railway lines and stations in India	Railway mapping
GADM Shapefiles	GADM.org	Geographic boundary files for India's 676 districts	District maps

3.2 Tools and Methods

I used the following tools to collect, clean, analyze, and visualize the data:

- Python (Pandas, GeoPandas, Matplotlib) — for data cleaning, transformation and exploratory visualization
- SQL (SQLite via Python) — for storing cleaned data in a relational database and running analytical queries
- Power BI — for building an interactive dashboard to present findings visually
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□ 3.3 Data Cleaning Process

The World Bank WDI dataset originally had 395,276 rows and 70 columns. Each row **represents** one country-indicator combination, and each column **represents** a year from 1960 to 2025. I filtered this down to only the **five** indicators relevant to my research question and kept only **the** years 2000 to 2022, where data quality was acceptable. The final cleaned dataset **contained** 15,641 rows.

For the **Indian** railway data, I loaded shapefiles containing 89,129 individual railway line segments and 8,654 station locations. For the district boundary data, I loaded **the** boundary polygons for all 676 districts in India.

Key Indicators Analyzed
GDP per capita (constant 2015 USD) — measures economic output per person Poverty headcount ratio at \$3/day — measures share of population in poverty Rail lines (total route-km) — measures railway infrastructure Access to electricity (% of population) — measures basic infrastructure access

4. Global Analysis — Findings

4.1 The Enormous GDP Gap Between Countries

The first thing to notice in the global GDP data set is the disparity. The average GDP per capita for the richest in the data set, Monaco, was \$179,782 between 2015 and 2022. For the poorest country, Burundi, the average was \$262. The difference is 682. Just to give an example of the disparity, in the average year, someone in Monaco would make what a person in Burundi would make in 730 days or approximately two years.

The top ten richest countries, which include Monaco, Liechtenstein, Luxembourg, Bermuda, the Isle of Man, Switzerland, the Cayman Islands, Norway, Macao, and Qatar, all have average GDPs of over \$60,000. The bottom ten poorest countries, which include Burundi, the Central African Republic, Madagascar, Somalia, the Democratic Republic of the Congo, Afghanistan, Niger, Malawi, Mozambique, and Liberia, all have GDPs of under \$650.

4.2 Electricity Access and Wealth — A Near-Perfect Pattern

One of the clearest findings in this dataset is the relationship between electricity access and economic status. When I filtered the data to look only at countries with 100% electricity access, every single one of them had a GDP per capita above USD 50,000. These countries include Monaco, Liechtenstein, Luxembourg, Switzerland, Norway, the United States, Singapore, and Australia.

In contrast, the countries with the highest poverty rates — Congo D.R. (82% poverty), Mozambique (79%), South Sudan (76%), Burundi (75%), and Malawi (73%) — all had electricity access below 36% of their population. South Sudan had the lowest percentage at 5.3%.

Finding 1
Every country with 100% electricity access has a per capita GDP above \$50,000. Every country with a poverty rate above 70% has an electricity access rate below 36%. This pattern was observed in all the cases in our dataset.

4.3 Rail Network and GDP

The relationship between rail network size and GDP per capita was very striking. Rail networks in the United States had the highest total length (158,380 km) and ranked 15th in rail network size among the world's largest economies. Rail networks existed in Germany (33,461 km) and had a GDP per capita of \$42,090. Similarly, Japan had rail networks of 20,113 km with an average GDP/capita of \$32,976.

However, since some of the countries that had the most amount of income based on GDP/capita (Monaco and Liechtenstein) and therefore had shorter rail networks because they are small, this illustrates the importance of considering both rail network sizes and country sizes.

Though it was imperfectly displayed, most countries have high levels of wealth relative to the amount spent on constructing railways throughout their entire rail systems.

4.4 India in the Global Context

Among its immediate neighbors, India's GDP per capita of \$1,844 (2015–2022 average) places it ahead of Pakistan (\$1,528) and Bangladesh (\$1,504), but far behind China (\$10,013) and the United States (\$59,739). India's per capita income is roughly 5.5 times lower than China's, even though both countries had similar economic sizes in the 1990s.

This comparison raises an important question for the Indian case study: within India, are the same infrastructure-related patterns visible at the district level?

5. India Case Study — Findings

5.1 India's GDP Growth Trajectory

India's GDP per capita growth story from 2000 to 2022 is genuinely impressive. In 2000, India's GDP per capita was \$756. By 2022, it had reached \$2,098, a growth of 177% over 22 years. This represents one of the fastest sustained periods of economic growth for any large country.

However, one important interruption is clearly visible in the data: in 2020, India's GDP per capita dropped from \$1,936 to \$1,806, a decline caused by the COVID-19 pandemic and the economic disruption it caused. The economy recovered strongly in 2021 (\$1,965) and continued growing in 2022 (\$2,098).

5.2 Railway Network Distribution Across India

The railway shapefile data, which contain 89,129 individual railway line segments, reveal a clear geographic pattern when plotted on a map of Indian districts. The railway network is heavily concentrated in the northern plains of Uttar Pradesh, Bihar, West Bengal, and Punjab, as well as in the peninsular corridor connecting Mumbai, Chennai, and Kolkata.

In contrast, northeastern states such as Meghalaya, Manipur, Mizoram, and Nagaland have very limited railway connectivity. These are also among the least economically developed regions of India, consistent with the infrastructure hypothesis.

Finding 2
India's railway network is highly unequal across the districts. The Indo-Gangetic Plain and coastal corridors are densely connected. Northeast India and many tribal districts have almost no railway access at all. This geographic inequality in infrastructure is closely related to economic inequality.

5.3 The Historical Infrastructure Argument for India

The current railway distribution in India is not random. Most of India's major railway lines were built by the British between 1853 and 1947, primarily to move cotton, coal, and troops efficiently, not necessarily to

develop local economies. However, the side effect of this infrastructure was that districts connected to the railway network gained access to markets, experienced population growth, and attracted economic activity.

Economist Dave Donaldson's research showed that when a railway line was built connecting a district in colonial India, the real agricultural income in that district increased by an average of 16%. The districts that were left out — often due to political decisions or geographic difficulty — never caught up to the same extent as the others.

This historical legacy arguably continues to shape regional inequality in India. Districts in Uttar Pradesh and Bihar that were connected to railways in the 1880s have larger cities, more industries, and better access to markets than comparable districts in Jharkhand or Northeast India that were connected much later or not connected at all.

6. Conclusion

This research was conducted to determine why certain areas stay impoverished. The research points to the lack of infrastructure specifically focusing on measurable means of transportation (electricity access, rail network length, etc.) as a primary factor by analyzing data from over 180 countries and 676 districts in India.

There is nearly a perfect correlation between having electrical access and GDP/capita (no nation with universal electricity access is currently classified as meeting the internationally defined criteria for being poor, and no nation with extreme poverty can be reasonably said to have significant access to electrical service). The correlation between connectivity by rail increases positively with increases in wealth; however, the relationship between the growth of GDP and rail line connectivity has been much stronger for larger developing nations than it has been for smaller developing nations.

The growth rate of GDP in India has been impressive since 2000 (177% over the last 20 years), despite the continuing high level of inequality between different Indian districts. The historical geographic distribution of rail transportation infrastructure in India as determined by the British during their colonial rule (especially during the last half of the 19th century) continues to substantially affect the extent to which districts in India are connected to regional/rural markets and their degree of economic activity; districts that are well connected to the railway system generally experience much higher levels of economic activity today.

From a geographic perspective, it is widely held that landlocked nations are at a disadvantage when it comes to economic development compared to nations with port access, and geographic barriers to building infrastructure will also impact economic growth due to the higher costs associated with constructing infrastructure in difficult geographic conditions. However, the data suggest that geography represents more of an initial constraint than it represents an everlasting fate (i.e., geographic disadvantages can be leveraged by investing in the necessary infrastructure needed to develop strong economies).

Summary of Key Findings
Finding 1: Electricity access is the strongest predictor of GDP per capita in this dataset. Finding 2: Countries with the highest poverty rates have electricity access below 36%. Finding 3: The rail network size positively correlates with GDP per capita across countries. Finding 4: India's GDP grew by 177% from 2000 to 2022, with a visible COVID dip in 2020. Finding 5: India's railway network is heavily concentrated in the northern and coastal corridors. Finding 6: Northeast India, with the lowest rail density, also shows the lowest connectivity to national markets.

6.1 Limitations

This study has several important limitations that I want to acknowledge. First, I was unable to access the complete Donaldson railway dataset **because of** access restrictions; **therefore**, the India district-level analysis is more descriptive than quantitative. Second, correlation does not mean causation; the fact that electricity access and GDP are correlated does not prove that electricity causes higher GDP. Third, the World Bank data have significant missing values for many countries and years, which may have affected some results.

6.2 Future Work

In future research, I would like to run a proper regression analysis to quantify how much of the variation in GDP per capita is explained by electricity access **compared to** geography. I would also like to include more India-specific variables, such as the NITI Aayog SDG index scores, to compare district-level development with railway access **more rigorously**.

7. References

- Acemoglu, D., & Robinson, J. A. (2012). *Why Nations Fail: The Origins of Power, Prosperity and Poverty*. Crown Publishers.
- Donaldson, D. (2018). Railroads of the Raj: Estimating the Impact of Transportation Infrastructure. *American Economic Review*, 108(4-5), 899–934.
- Gallup, J. L., Sachs, J. D., & Mellinger, A. D. (1999). Geography and Economic Development. *International Regional Science Review*, 22(2), 179–232.
- Sachs, J. D. (2003). *Institutions Do not Rule: Direct Effects of Geography on Per Capita Income*. NBER Working Paper No. 9490.
- World Bank. (2024). *World Development Indicators (WDI)*. Washington, D.C.: The World Bank Group. Retrieved from <https://databank.worldbank.org/source/world-development-indicators>
- Census of India. (2011). *Housing and Primary Census Abstracts*. Office of the Registrar General and Census Commissioner, India.
- OpenStreetMap Contributors. (2025). *India Railways Dataset*. Humanitarian Data Exchange. Retrieved from https://data.humdata.org/dataset/hotosm_ind_railways
- GADM. (2024). *Global Administrative Areas — India Level 2*. Retrieved from <https://gadm.org>